Instrumentation Calibration Workbench

Python lab instrumentation · telemetry sync · calibration lineage · LunarRego plasma diagnostics

Portfolio showcase PDF

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Public repository with clickable documentation links
Includes DAQ/TDK logging architecture, calibration artifacts, and LP/EP/RPA analysis

https://github.com/harshddes/instrumentation-calibration-workbench

Contents

Click blue URLs throughout this PDF to open GitHub documentation.

1. Project overview

This repository is a curated public showcase of research-engineering software for laboratory instrumentation. It shows how Python can coordinate independent instruments, merge their telemetry with explicit freshness metadata, preserve calibration lineage in versioned artifacts, and operate plasma diagnostics for a lunar-regolith electric-field experiment.

Core idea — DAQ + TDK

A DAQ logger records measurement channels. A power-supply logger publishes the latest telemetry snapshot as JSON. Each DAQ row records whether that telemetry was fresh, stale, or missing. A calibration package then maps raw DAQ current readings onto reference-scale values using a versioned JSON artifact.

Extended path — LunarRego

Beyond DAQ/TDK, the project includes a LunarRego operator GUI for Langmuir Probe (LP), Emissive Probe (EP), and Retarding Potential Analyzer (RPA). Inside the vacuum chamber, a bias electrode is driven positive or negative to generate the desired electric field while lunar-regolith simulant lofting / charging is observed. LP and EP provide plasma-potential markers; early RPA collector sweeps are retained as rudimentary pipeline results only (not claimed as plasma potential).

Vacuum system status (current)

The chamber vacuum train is a roughing pump combined with a CTI cryopump. There is currently a roughing-pump problem, so the team is troubleshooting the vacuum system while preparing for additional diagnostic runs. In parallel, work is underway to automate the automatic valve controller (AVC) that sequences the roughing / cryopump path.

What this shows

- Multi-instrument logging with an explicit JSON snapshot bridge
- Keithley DAQ scan logic with CSV output and TDK telemetry columns
- TDK Lambda telemetry/control with GUI operation and safety-oriented state handling
- LunarRego LP / EP / RPA GUI with hardware-map role assignment and Keithley backends
- Public I–V review for approved LP / EP / RPA CSVs, including dI/dV markers
- Calibration generator with source hashes, cleaning rules, coefficients, and review plots
- Reusable calibration library for scripts, notebooks, and downstream analysis

Documentation links

Repository README → https://github.com/harshddes/instrumentation-calibration-workbench/blob/main/README.md
Documentation index → https://github.com/harshddes/instrumentation-calibration-workbench/blob/main/docs/README.md
Architecture / data flow → https://github.com/harshddes/instrumentation-calibration-workbench/blob/main/docs/architecture/da
Calibration methodology → https://github.com/harshddes/instrumentation-calibration-workbench/blob/main/docs/calibration/me
Plasma diagnostics methodology → https://github.com/harshddes/instrumentation-calibration-workbench/blob/main/docs/plasm
Workflow runbooks → https://github.com/harshddes/instrumentation-calibration-workbench/blob/main/docs/workflows/runbook
Reproducibility boundary → https://github.com/harshddes/instrumentation-calibration-workbench/blob/main/docs/reproducibility
Project Wiki → https://github.com/harshddes/instrumentation-calibration-workbench/wiki
Wiki source: Plasma Diagnostics → https://github.com/harshddes/instrumentation-calibration-workbench/blob/main/wiki/Plasm

2. Architecture: DAQ ↔ TDK bridge

Two instruments produce related data without a shared process clock. The implemented solution uses a small JSON snapshot as the bridge between TDK telemetry and DAQ rows. Timing uncertainty stays visible as freshness status instead of being hidden.

Data flow

- TDK logger → TDK CSV and published snapshot JSON
- DAQ logger → channel readings
- Snapshot → freshness status (fresh / stale / missing / invalid)
- Merged DAQ CSV carries readings + TDK columns + freshness
- Merged CSV → calibration generator → versioned artifact → calibration library
- Merged CSV → dashboard / analysis

Snapshot contract (example fields)

ps_1_voltage, ps_1_current, ps_1_output_state, ps_2_voltage, ps_2_current, ps_2_output_state, plus timestamp / sequence / published_at metadata.

Open snapshot helper source → instrumentation/snapshot.py
Open architecture doc → docs/architecture/dataflow.md

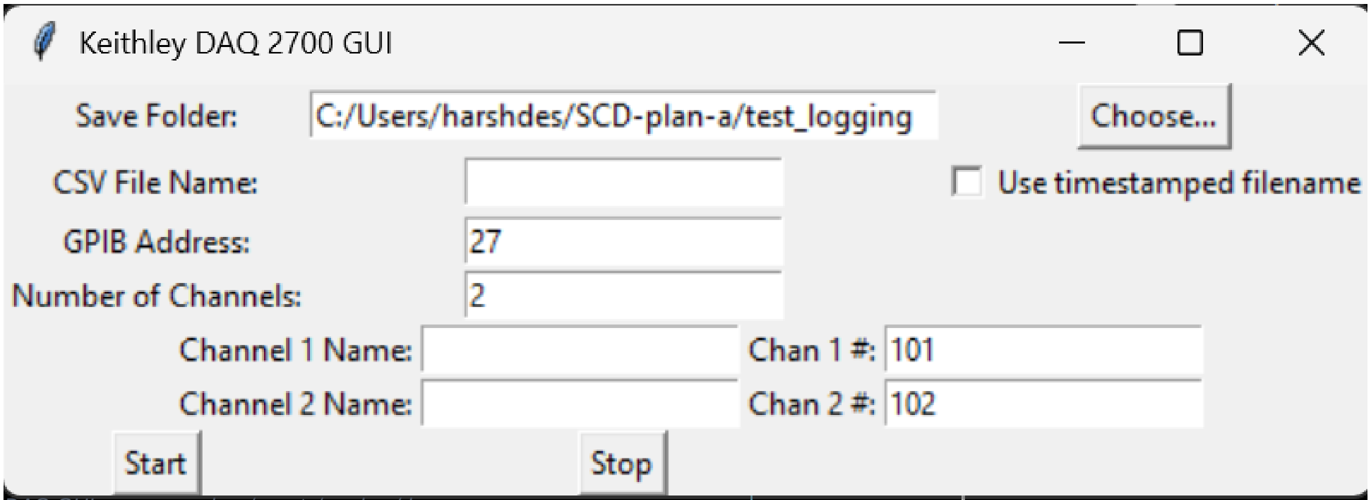
LunarRego diagnostics path

- Hardware Map JSON assigns instrument / panel / GPIB / board per LP, EP, RPA plate role
- GUI acquires LP CSV and RPA combined collector CSV; EP sheet is reviewed from approved export
- analyze_iv_curves computes I–V and dI/dV; LP/EP markers are plasma-potential related
- RPA public review is rudimentary only — not labeled plasma potential
- Vacuum: roughing + CTI cryopump; roughing-pump troubleshooting and AVC automation in progress

3. Operator GUIs

Keithley DAQ 2700

Tk operator surface for multi-channel DAQ logging with folder/CSV controls and merged TDK telemetry columns.

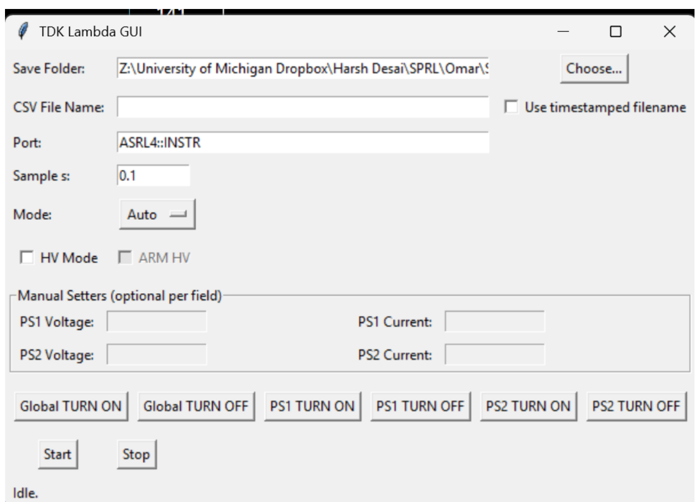


DAQ GUI — source: docs/assets/readme/daq.png

Run: python -m instrumentation.daq.GUI_DAQ2700

TDK Lambda

Telemetry/control GUI for supply logging, snapshot publish, and operator-facing manual/auto controls with safety-oriented state handling.

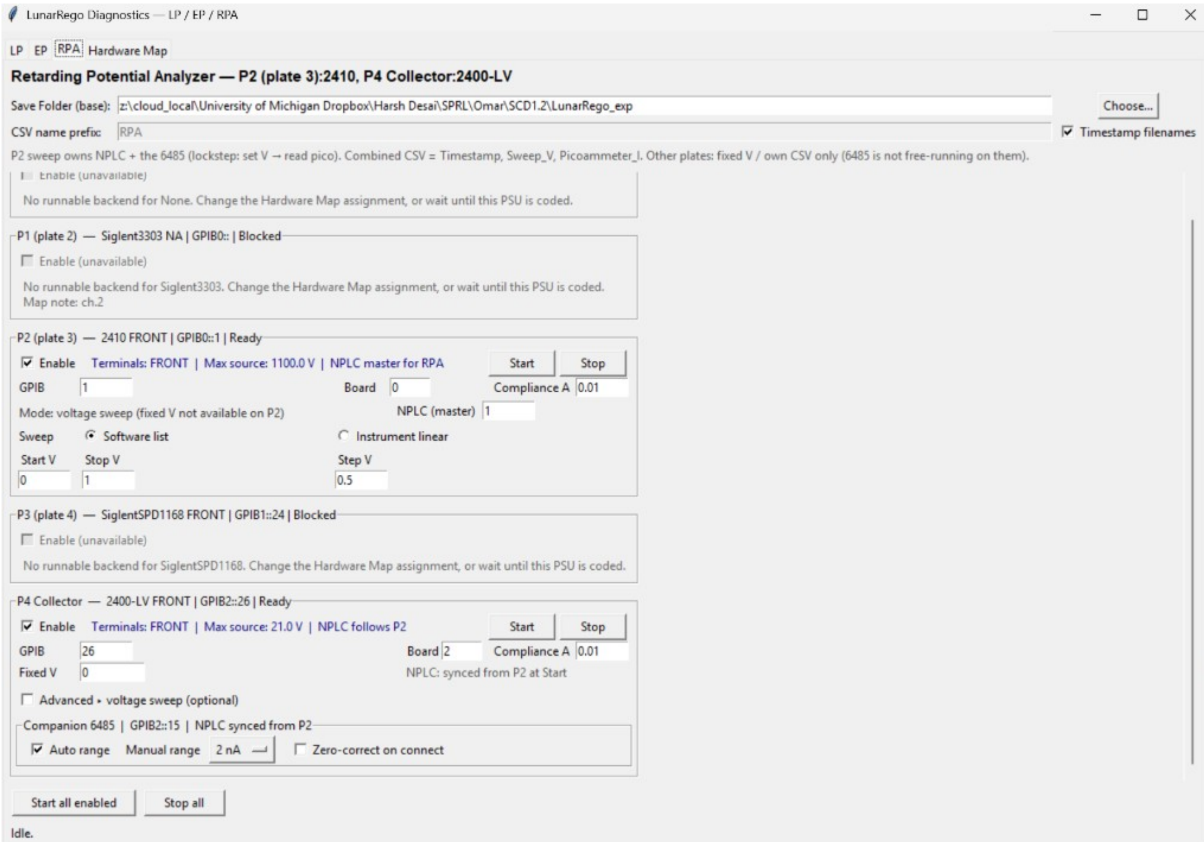


TDK GUI — source: docs/assets/readme/tdk.png

Run: python -m instrumentation.tdk.GUI_TDK

3b. LunarRego diagnostics GUI

LP / EP / RPA operator GUI with Hardware Map roles. On the RPA tab, P2 (Keithley 2410) owns the retarding-voltage sweep and NPLC master timing; P4 collector (Keithley 2400-LV) holds collector bias while a Keithley 6485 picoammeter records collector current in lockstep (set V → read pico). Combined CSV columns: Timestamp, Sweep_V, Picoammeter_I.



LunarRego GUI (RPA tab) — source: docs/assets/readme/lunar-rego-gui.png

Run: python -m instrumentation.lunar_rego.GUI_LunarRego

4. Chamber experiment and bias electrode

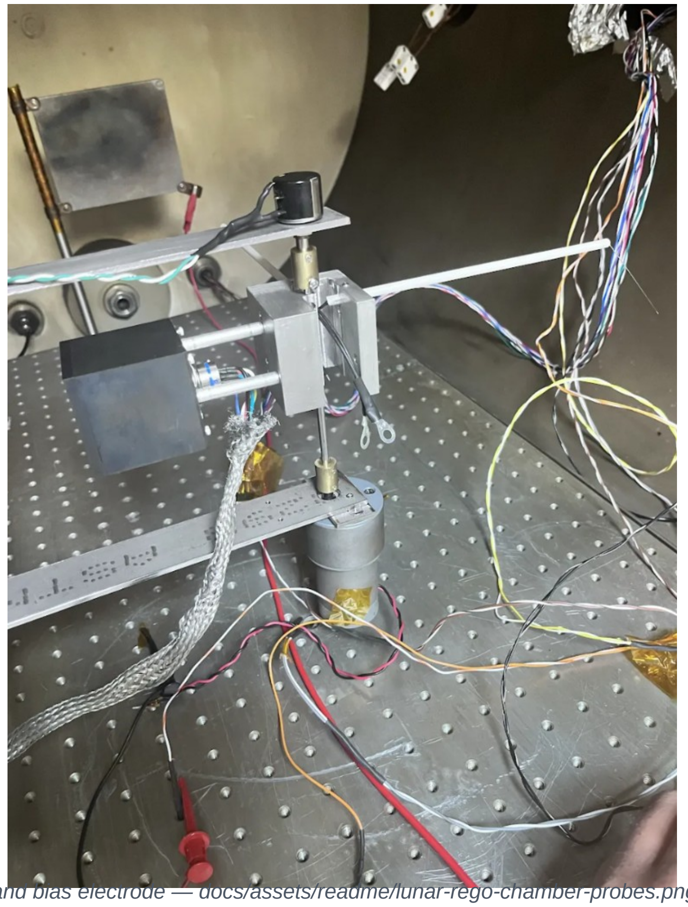
LunarRego is a vacuum-chamber campaign that couples plasma diagnostics to lunar-regolith-simulant lofting under a controlled electric field.

Bias electrode = field actuator

The square metallic plate mounted on a vertical rod is the bias electrode. It is driven to a chosen positive or negative potential to generate the desired electric field. Dust / regolith simulant response (charging, lofting, transport) is then interpreted against the local plasma state measured by LP and EP — not against an assumed plasma potential. Fuller RPA campaigns are planned after vacuum recovery.

Vacuum train and current status

The chamber uses a roughing pump + CTI cryopump combination. A roughing-pump issue is under active troubleshooting, which pauses additional diagnostic runs for now. In parallel, the automatic valve controller (AVC) is being automated to sequence the roughing / cryopump path more reliably for future campaigns.



Vacuum-chamber probe assembly and bias electrode — docs/assets/readme/lunar-rego-chamber-probes.png

- Horizontal probe / central stack: plasma-diagnostics path into the measurement volume
- Left square plate: bias electrode for +/– electric-field control
- Feedthrough wiring connects SMUs / picoammeter / supplies outside the chamber

5. Plasma diagnostics — LP / EP / RPA

Probe roles

Probe	Measures	Public marker
LP	I vs probe bias	V* = dI/dV peak; V_f at I≈0
EP	Floating V vs emission	Vp ≈ high-emission asymptote
RPA	Collector I vs retarding V	Rudimentary dI/dV only (not Vp)

RPA acquisition contract

- P2 = retarding / discriminator plate (software sweep, NPLC master)
- P4 = collector bias on 2400-LV; 6485 companion reads picoammeter current
- Lockstep: for each Sweep_V, set P2 then read 6485
- Other plates may hold fixed bias / own CSV; 6485 is not free-running on them

Approved public datasets only

LP_07302026_140808.csv
RPA_combined_07302026_170652.csv
EP_PlasmaDiagnostics_exp.csv

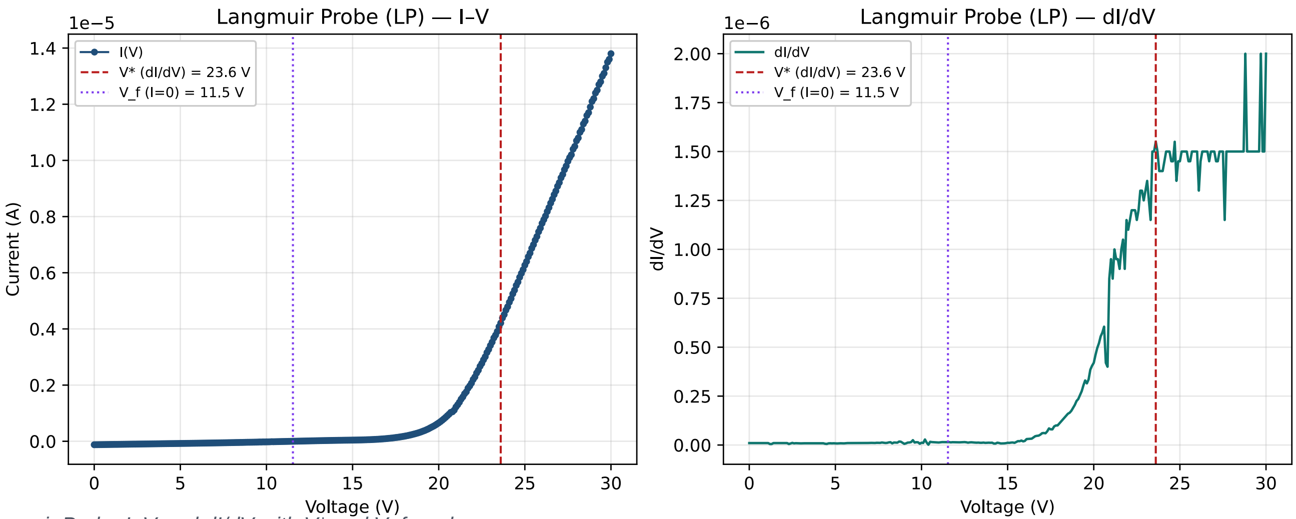
Regenerated review markers

Diagnostic	Estimate	Notes
LP	V* ≈ 23.6 V; V_f ≈ 11.5 V	Plasma-potential related (dI/dV)
EP	Vp ≈ 21.3 V	High-emission floating asymptote
RPA	dI/dV feature ≈ 21.5 V	Rudimentary only — NOT plasma potential

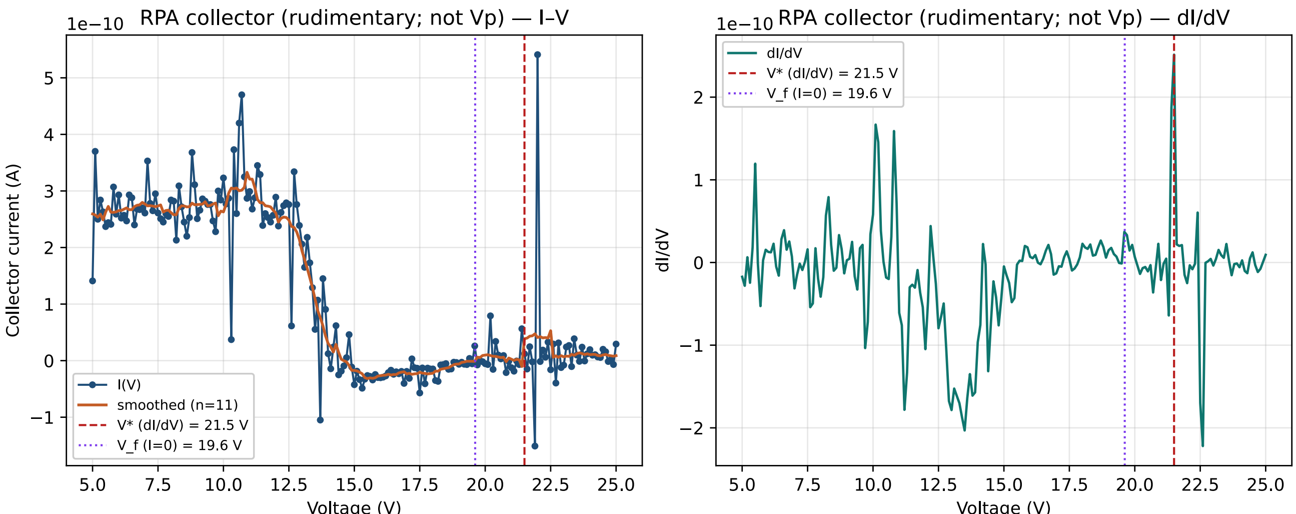
RPA is shown only as an early pipeline example while the vacuum system (roughing pump + CTI cryopump) is troubleshot and AVC automation continues. More RPA results are planned after vacuum recovery.

Methodology → docs/plasma_diagnostics/methodology.md
Regenerate plots → instrumentation/lunar_rego/analyze_iv_curves.py

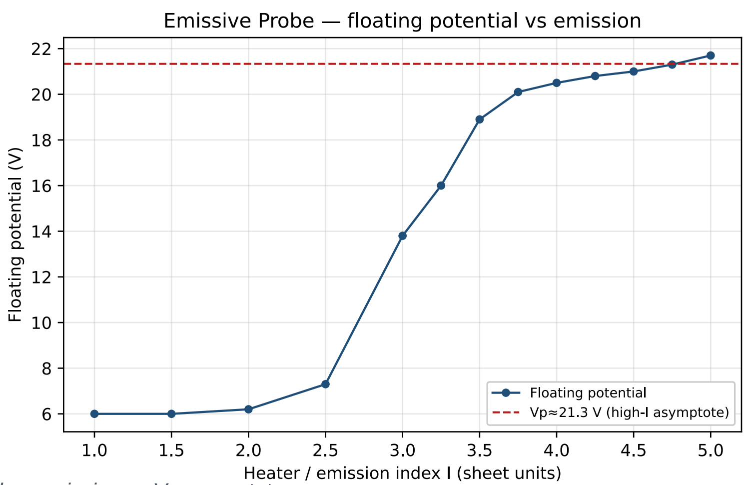
5b. Review plots



Langmuir Probe I–V and dI/dV with V and V_f markers*



RPA collector I–V and dI/dV — rudimentary early result (not plasma potential)



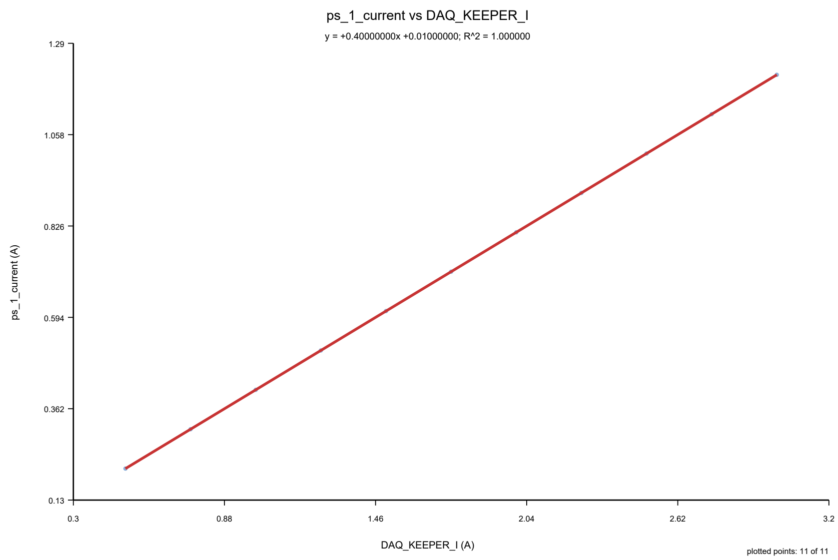
Emissive Probe floating potential vs emission → Vp asymptote

6. Calibration lineage demo

The public calibration demo is generated from synthetic data so reviewers can inspect the full lineage without private experiment archives: source CSV, cleaned CSV, rejected rows, JSON metadata (hashes, coefficients, fit quality), and an SVG review plot.

Demo model

ps_1_current = 0.4 * DAQ_KEEPER_I + 0.01



Synthetic keeper-current calibration review plot

Artifact JSON → calibration_process/artifacts/demo_keeper_current_linear.json
Generator → calibration_process/generate_keeper_current_calibration.py
Runtime library → calibration_process/library.py
Methodology → docs/calibration/methodology.md

7. Repository map and quick start

Key paths

instrumentation/daq/
Keithley DAQ class + GUI

instrumentation/tdk/
TDK telemetry/control + GUI

instrumentation/lunar_rego/
LP/EP/RPA GUI, backends, IV review

instrumentation/snapshot.py
TDK↔DAQ snapshot contract

calibration_process/
Versioned calibration generator/library

examples/plasma_diagnostics/
Approved LP/EP/RPA CSVs only

docs/
Architecture, calibration, plasma, runbooks

wiki/
GitHub Wiki source pages

SCD_3D_AI_Lab/
Streamlit CSV exploration dashboard

code_xray/
Static/dynamic DAQ tracing experiment

Quick start

- python -m venv .venv
- .\.venv\Scripts\Activate.ps1
- python -m pip install -r requirements.txt
- python -m calibration_process.generate_keeper_current_calibration
- python -m instrumentation.daq.GUI_DAQ2700
- python -m instrumentation.tdk.GUI_TDK
- python -m instrumentation.lunar_rego.GUI_LunarRego
- python -m instrumentation.lunar_rego.analyze_iv_curves

Runbooks → docs/workflows/runbooks.md

Requirements → requirements.txt

Public-release boundary

This repository is curated from a larger working lab tree. Large experiment archives, scratch folders, and private editor metadata are excluded. For LunarRego, only the three approved LP / EP / RPA CSVs are public.

Reproducibility notes → docs/reproducibility.md

Repository home

https://github.com/harshddes/instrumentation-calibration-workbench